

Customer Retention & Churn Analysis Dashboard

Overview

This project presents an interactive **Customer Retention & Churn Analysis Dashboard** developed using **Microsoft Power BI**. The dashboard analyzes customer churn behavior, retention trends, and key business metrics to help organizations understand why customers leave and identify strategies to improve customer retention.

This project was completed as part of the **Future Interns – Data Science & Analytics Internship (Task 2)**.

Problem Statement

Customer churn directly impacts business growth and revenue. The objective of this project is to analyze customer behavior, identify churn patterns, and provide actionable business recommendations to improve customer retention.

Objectives

- Analyze customer churn rate.
 - Measure customer retention performance.
 - Identify high-risk customer segments.
 - Study customer tenure and monthly charges.
 - Compare churn across contract types and internet services.
 - Generate business insights and recommendations.
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Tools Used

- Microsoft Power BI
 - Power Query
 - DAX (Data Analysis Expressions)
 - Microsoft Excel
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Dataset

Dataset: Telco Customer Churn Dataset

The dataset contains customer demographic information, subscription details, contract information, billing details, and churn status.

Dashboard Features

KPI Cards

- Total Customers
- Active Customers
- Churn Customers
- Churn Rate
- Average Customer Tenure
- Average Monthly Charges

Visualizations

- Customer Churn Distribution
- Churn by Contract Type
- Churn by Internet Service
- Monthly Charges vs Churn
- Churn by Payment Method
- Customer Tenure vs Churn

Interactive Filters

- Gender
- Contract Type
- Payment Method
- Internet Service

Key Insights

- Overall customer churn rate is **26.54%**.
- Month-to-Month contract customers have the highest churn.
- Fiber Optic users show higher churn than DSL users.
- Electronic Check customers are more likely to leave.
- Customers with shorter tenure are at greater risk of churn.

Business Recommendations

- Promote yearly and two-year contracts using discounts.
 - Improve onboarding for new customers.
 - Enhance customer support for Fiber Optic users.
 - Provide personalized retention offers to high-risk customers.
 - Implement loyalty programs for long-term customers.
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Project Files

```
Customer-Retention-Churn-Analysis/  
|  
├─ Customer_Retention_Churn_Analysis.pbix  
├─ WA_Fn-UseC_-Telco-Customer-Churn.csv  
├─ Customer_Retention_Dashboard.png  
├─ Customer_Retention_Churn_Analysis_Report.pdf  
└─ README.md
```

Skills Demonstrated

- Data Cleaning
- Data Transformation
- DAX Calculations
- Customer Churn Analysis
- Business Intelligence
- Dashboard Development
- Data Visualization
- Business Insight Generation

Dashboard Preview

Add your dashboard screenshot here after uploading it to GitHub.

Author

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Future Interns – Data Science & Analytics Internship

Power BI | Data Analytics | Business Intelligence